

DIPESH GHIMIRE

Computer Science & IT Student | Python & Django Developer | Aspiring AI/ML Engineer

Kathmandu, Nepal | 9810534340 | ghimiredipesh33@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

8th-semester B.Sc. CSIT student with practical experience developing a team-based web platform using Python and Django, and an end-to-end machine learning project for early sepsis prediction using XGBoost and Streamlit. Comfortable with Python programming, database-driven application development, and integrating external services such as OpenStreetMap. Currently developing foundational skills in Data Science and Machine Learning, including regression, classification, statistics, NumPy, Pandas, and Matplotlib, reinforced by hands-on Kaggle courses and multiple job simulations. Interested in applying software engineering skills to AI/ML, NLP, computer vision, and intelligent document-processing systems.

EDUCATION

B.Sc. Computer Science and Information Technology (B.Sc. CSIT) | *Nihareeka College, affiliated with Tribhuvan University*

Expected Graduation: 2083

PROJECTS

Sepsis Prediction Using Machine Learning — Internship Project | *Python, XGBoost, Scikit-learn, SHAP, Streamlit*

- Built an end-to-end ML pipeline for early sepsis prediction on the PhysioNet/CinC Challenge 2019 ICU dataset (1.55M observations, 40,336 patients, ~1.8% positive rate).
- Performed exploratory analysis of class imbalance, missingness (up to 99% in lab values), correlations, and temporal patterns.
- Prevented data leakage with patient-level train/validation/test splitting (70/10/20) and patient-wise forward filling of clinical measurements.
- Engineered five physiological domain scores from reference-range deviations with training-derived weights, reducing the clinical feature space to 10 model features.
- Trained an XGBoost classifier with class-imbalance weighting and compared it against Logistic Regression, Random Forest, HistGradientBoosting, and Linear SVM baselines.
- Selected the decision threshold on validation data only; XGBoost achieved the highest held-out test ROC-AUC (0.791) and PR-AUC (0.093) of the five models evaluated.
- Analyzed model behavior using feature importance, permutation importance, and SHAP; documented each stage in six notebooks with written summaries.
- Developed a Streamlit web app that reproduces the feature-engineering pipeline and returns a sepsis probability and prediction from the saved model.

GitHub: github.com/DipeshGhimire33/Intern_project

Tourist Platform — Team Project | *Python, Django, SQLite, HTML, CSS, JavaScript, OpenStreetMap*

- Developed a web platform with a team of 3 to simplify travel planning by bringing together services such as vehicles, equipment, guides, and hotels.
- Built the application using Python and Django, gaining practical experience in backend development and database-driven application design.
- Implemented Django authentication for user account and access management.
- Integrated OpenStreetMap to support route and location-related functionality.
- Worked extensively with SQLite and Django's database models during development.
- Addressed database migration and schema-management challenges when modifying application models, developing a practical understanding of how changes to data models affect an application.
- Collaborated with teammates throughout development and worked with a shared project codebase.

GitHub: github.com/DipeshGhimire33/Tourist_travels_7th_sem_project

TECHNICAL SKILLS

Programming: Python, C, C++

Web Development: Django, HTML, CSS, JavaScript

Data & Scientific Computing: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Regression, Classification, XGBoost, Scikit-learn, SHAP, Machine Learning Fundamentals

Application Development: Streamlit

Database: SQLite, SQL fundamentals

Other: Git/GitHub, Jupyter Notebook, REST/API concepts, OpenStreetMap integration

DATA SCIENCE & MACHINE LEARNING

Currently completing a 100-day Data Science & Machine Learning program

- Building foundational knowledge in machine learning and data analysis.
- Familiar with regression and classification concepts, applied in a hands-on imbalanced-classification project.
- Developing practical experience with NumPy, Pandas, and Matplotlib.
- Academic background in statistics through B.Sc. CSIT coursework.
- Currently expanding knowledge toward NLP, AI/ML applications, and real-world machine learning workflows.

GitHub: github.com/DipeshGhimire33/Data-Science-and-Machine-learning

CERTIFICATIONS

Kaggle Learn — Certificates of Completion

- **Intro to Programming** — Kaggle • September 1, 2026
- **Python** — Kaggle • September 2, 2026
- **Pandas** — Kaggle • September 3, 2026
- **Data Visualization** — Kaggle • September 3, 2026
- **Data Cleaning** — Kaggle • September 4, 2026
- **Feature Engineering** — Kaggle • September 6, 2026

Internship

- **Certificate of Experience — Machine Learning Data Analyst, Agribusiness** — YuvaIntern • September 7, 2026

Forage Job Simulations

- **GenAI Powered Data Analytics Job Simulation** — Tata (via Forage) • September 19, 2026
- **Data Analytics Job Simulation** — Deloitte (via Forage) • September 19, 2026
- **Data Labeling Job Simulation** — Forage • September 20, 2026
- **Data Science Job Simulation** — British Airways (via Forage) • September 20, 2026
- **Introduction to Data for Decision Makers Job Simulation** — BCG X (via Forage) • September 21, 2026

SOFT SKILLS

- **Team Collaboration:** Experience working in a 3-member development team and contributing to a shared codebase.
- **Problem Solving:** Hands-on experience troubleshooting database and schema-change issues during application development.
- **Adaptability & Learning:** Actively expanding from software development into Data Science and Machine Learning.
- **Communication:** Comfortable discussing technical challenges, asking questions, and working through problems collaboratively.
- **Curiosity:** Strong interest in understanding and applying emerging AI/ML technologies to practical problems.